Expt 4	Design of Low Pass Filter using Windowing Techniques
Date:	4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

Code:

```
clc;
clear;
close all;

%-------------------------------------------------------
% EX NO : 4
% TITLE : Design and Analysis of FIR Low-Pass Filter
%         Using Different Window Techniques
%-------------------------------------------------------

% Filter Specifications
fp = 1000;        % Passband Cutoff Frequency (Hz)
Fs = 4000;        % Sampling Frequency (Hz)
N  = 50;          % Filter Order (Even Number Preferred)

% Normalized Cutoff Frequency
Wc = fp / (Fs/2);

%-------------------------------------------------------
% FIR Low-Pass Filter Using Rectangular Window
%-------------------------------------------------------
b_rect = fir1(N, Wc, 'low', rectwin(N+1));

%-------------------------------------------------------
% FIR Low-Pass Filter Using Hamming Window
%-------------------------------------------------------
b_hamm = fir1(N, Wc, 'low', hamming(N+1));

%-------------------------------------------------------
% FIR Low-Pass Filter Using Hanning Window
%-------------------------------------------------------
b_hann = fir1(N, Wc, 'low', hann(N+1));

%-------------------------------------------------------
% Display Filter Coefficients
%-------------------------------------------------------
fprintf('------------------------------------------\n');
fprintf('Rectangular Window Coefficients\n');
fprintf('------------------------------------------\n');
disp(b_rect');
```

```
fprintf('-----------------------------------------\n');
fprintf('Hamming Window Coefficients\n');
fprintf('-----------------------------------------\n');
disp(b_hamm');

fprintf('-----------------------------------------\n');
fprintf('Hanning Window Coefficients\n');
fprintf('-----------------------------------------\n');
disp(b_hann');

%-----------------------------------------------------
% Frequency Response - Rectangular Window
%-----------------------------------------------------
figure;
freqz(b_rect, 1, 1024, Fs);
title('FIR Low-Pass Filter Using Rectangular Window');

%-----------------------------------------------------
% Frequency Response - Hamming Window
%-----------------------------------------------------
figure;
freqz(b_hamm, 1, 1024, Fs);
title('FIR Low-Pass Filter Using Hamming Window');

%-----------------------------------------------------
% Frequency Response - Hanning Window
%-----------------------------------------------------
figure;
freqz(b_hann, 1, 1024, Fs);
title('FIR Low-Pass Filter Using Hanning Window');

%-----------------------------------------------------
% Impulse Response Comparison
%-----------------------------------------------------
figure;

subplot(3,1,1);
impz(b_rect,1);
title('Impulse Response - Rectangular Window');

subplot(3,1,2);
impz(b_hamm,1);
title('Impulse Response - Hamming Window');

subplot(3,1,3);
impz(b_hann,1);
title('Impulse Response - Hanning Window');

%-----------------------------------------------------
```

```
% Pole-Zero Plots
%--------------------------------------------------------
figure;

subplot(1,3,1);
zplane(b_rect,1);
title('Rectangular Window');

subplot(1,3,2);
zplane(b_hamm,1);
title('Hamming Window');

subplot(1,3,3);
zplane(b_hann,1);
title('Hanning Window');

%--------------------------------------------------------
% Display Filter Details
%--------------------------------------------------------
fprintf('------------------------------------------\n');
fprintf('FIR Low-Pass Filter Specifications\n');
fprintf('------------------------------------------\n');
fprintf('Filter Order              = %d\n', N);
fprintf('Sampling Frequency (Fs)   = %d Hz\n', Fs);
fprintf('Cutoff Frequency (Fp)     = %d Hz\n', fp);
fprintf('Normalized Cutoff (Wc)    = %.4f\n', Wc);
fprintf('------------------------------------------\n');
```

Output:

Fig 1:

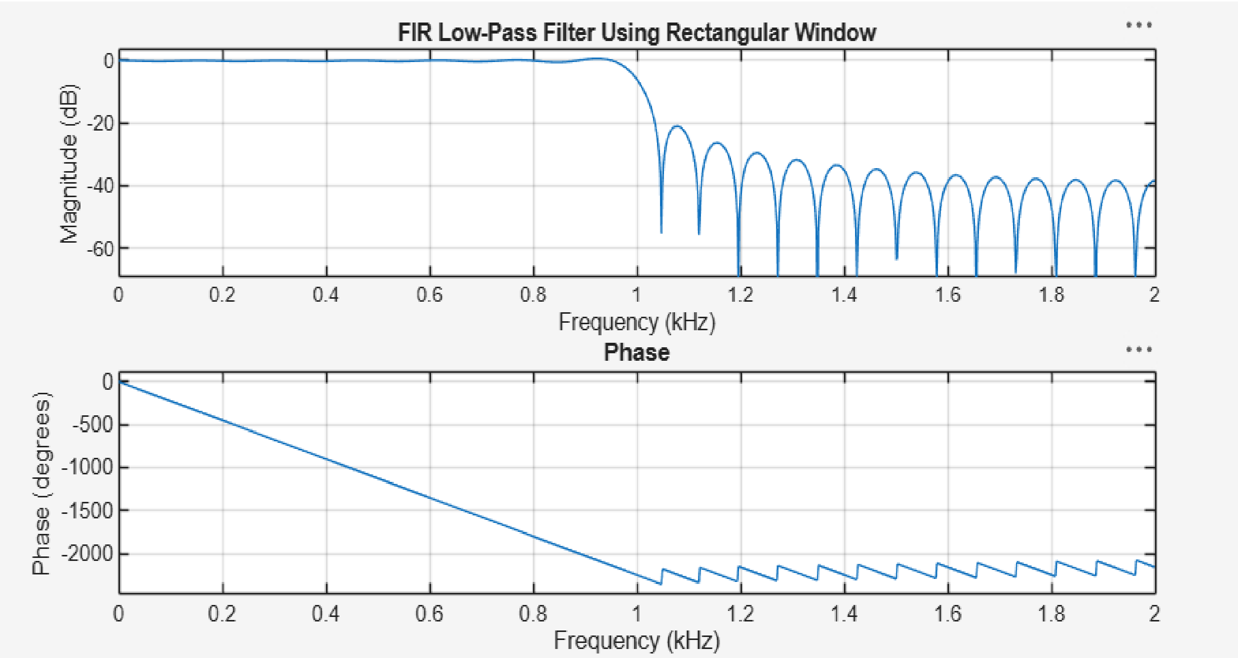


Fig 2:

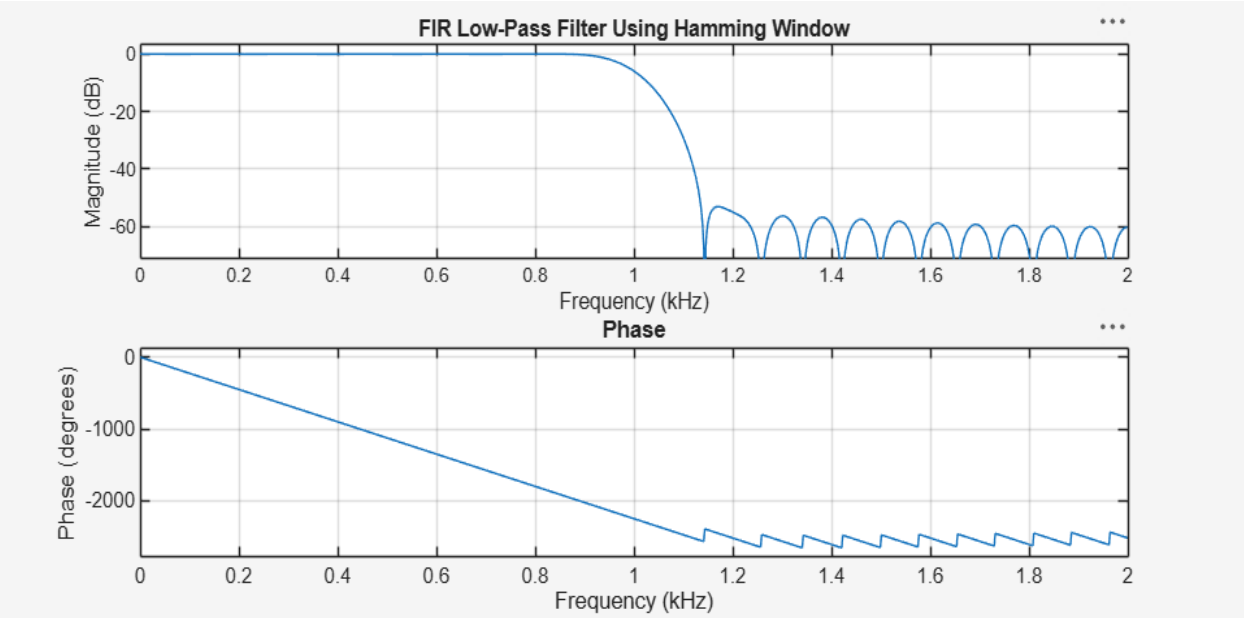


Fig 3:

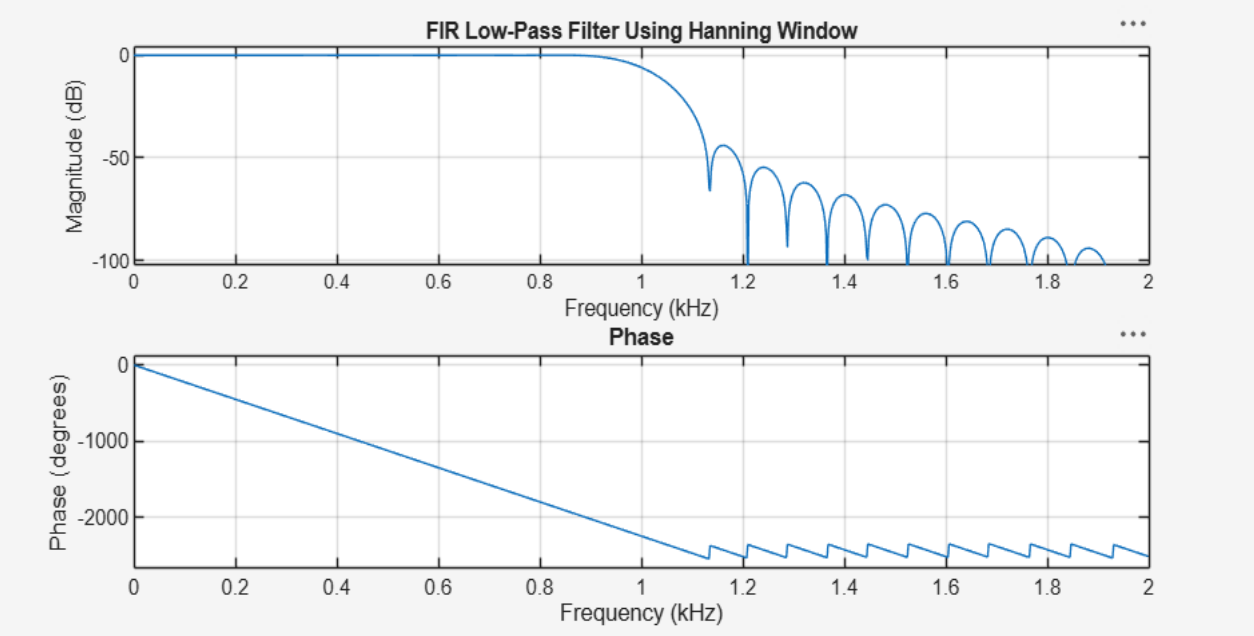
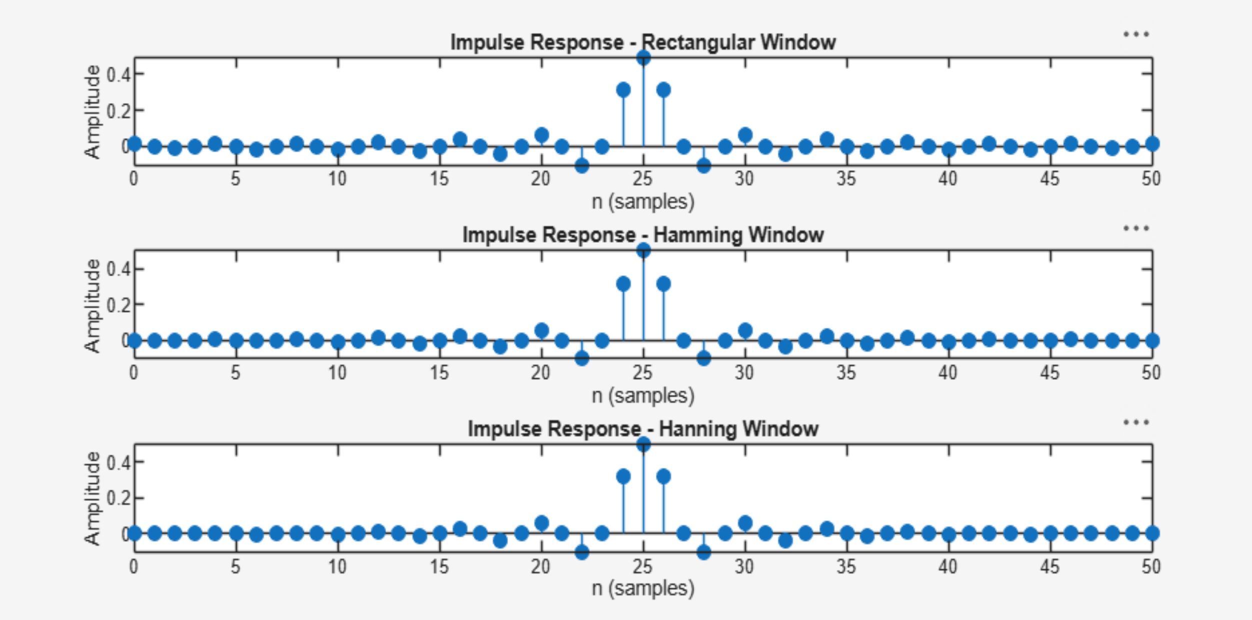
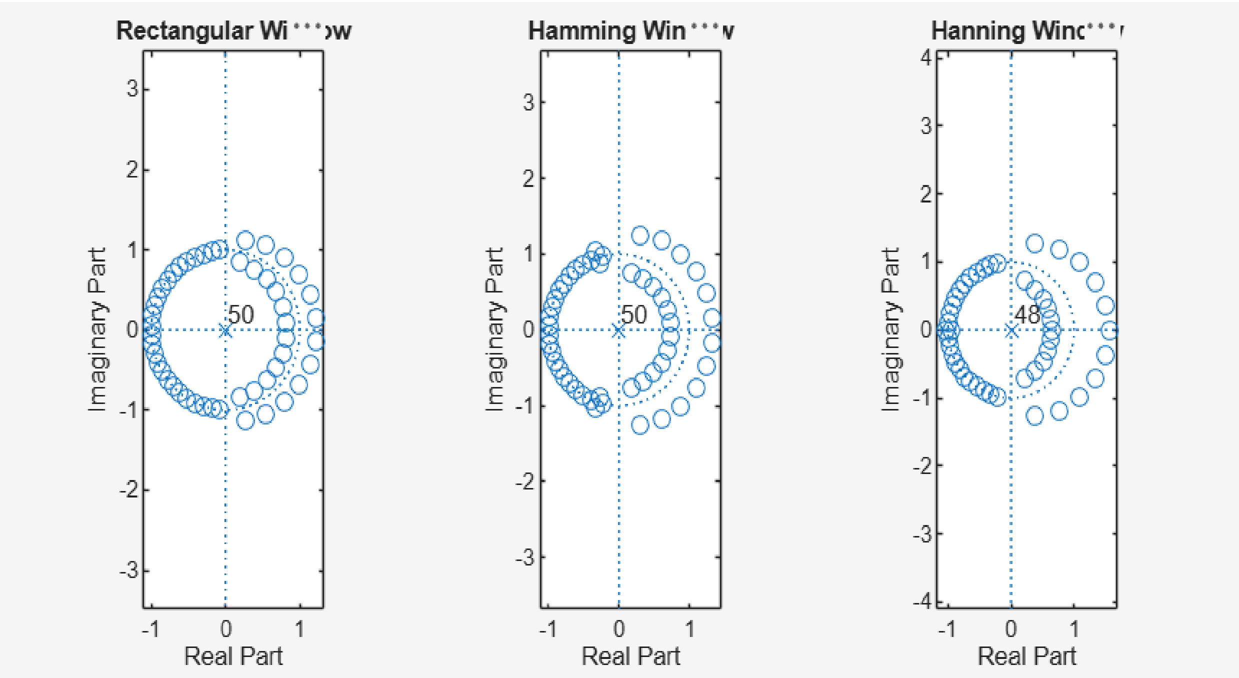


Fig 4:



Impulse Response - Rectangular Window
Amplitude
n (samples)
Impulse Response - Hamming Window
Amplitude
n (samples)
Impulse Response - Hanning Window
Amplitude
n (samples)

Fig 5:



Imaginary Part
Real Part
50
50
48

Expt 4	Design of Low Pass Filter using Windowing Techniques
Date:	4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

Code:

```
clc;
clear;
close all;

%% ==========================
% FIR Low-Pass Filter Design
% Using Hamming Window
%% ==========================

% Input Parameters
fp = 1000;        % Passband cutoff frequency (Hz)
Fs = 4000;        % Sampling frequency (Hz)
N  = 50;          % Filter order

% Normalized cutoff frequency
Wn = fp/(Fs/2);

%% Design FIR Low-Pass Filter
b = fir1(N, Wn, 'low', hamming(N+1));

%% Display Filter Information
disp('=====================================');
disp(' FIR Low-Pass Filter (Hamming Window)');
disp('=====================================');
fprintf('Filter Order          : %d\n', N);
fprintf('Sampling Frequency    : %d Hz\n', Fs);
fprintf('Cutoff Frequency      : %d Hz\n', fp);
fprintf('Normalized Cutoff     : %.2f\n\n', Wn);

disp('Filter Coefficients:');
disp(b');

%% Plot Frequency Response
figure;
freqz(b,1,1024,Fs);
title('Frequency Response of FIR Low-Pass Filter (Hamming Window)');

%% Plot Impulse Response
figure;
stem(0:N,b,'filled');
grid on;
```

```
xlabel('Sample Number');
ylabel('Amplitude');
title('Impulse Response of FIR Low-Pass Filter');

%% Plot Magnitude Response
[H,f] = freqz(b,1,1024,Fs);

figure;
plot(f,abs(H),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Magnitude Response');
xlim([0 Fs/2]);
```

Output:

Fig 1:

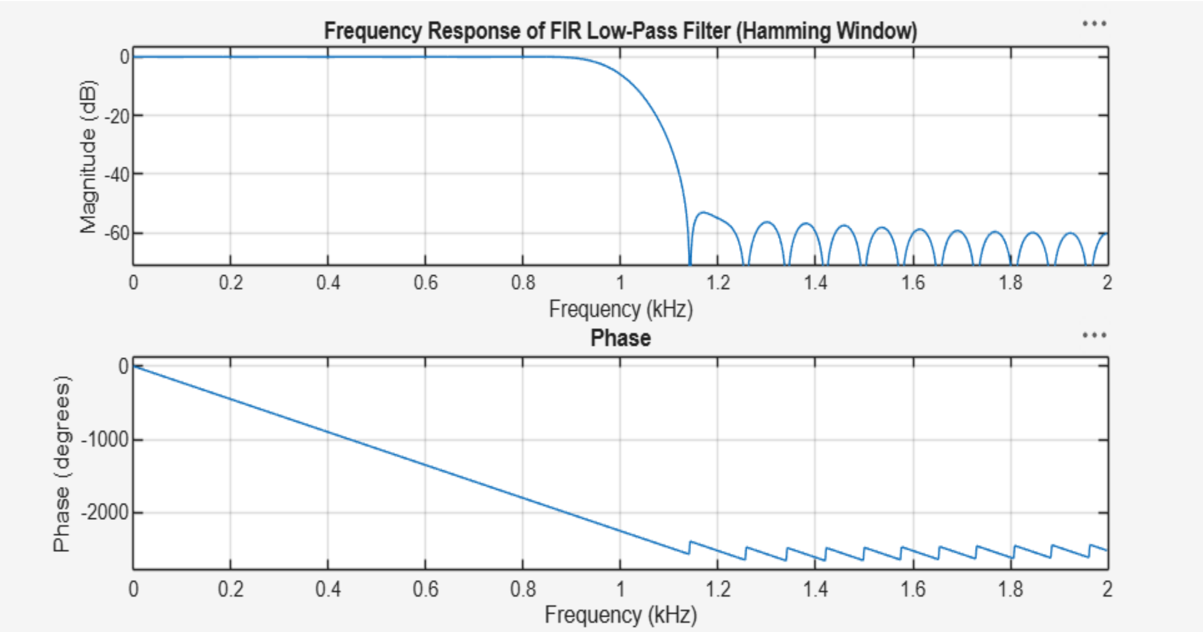


Fig 2:

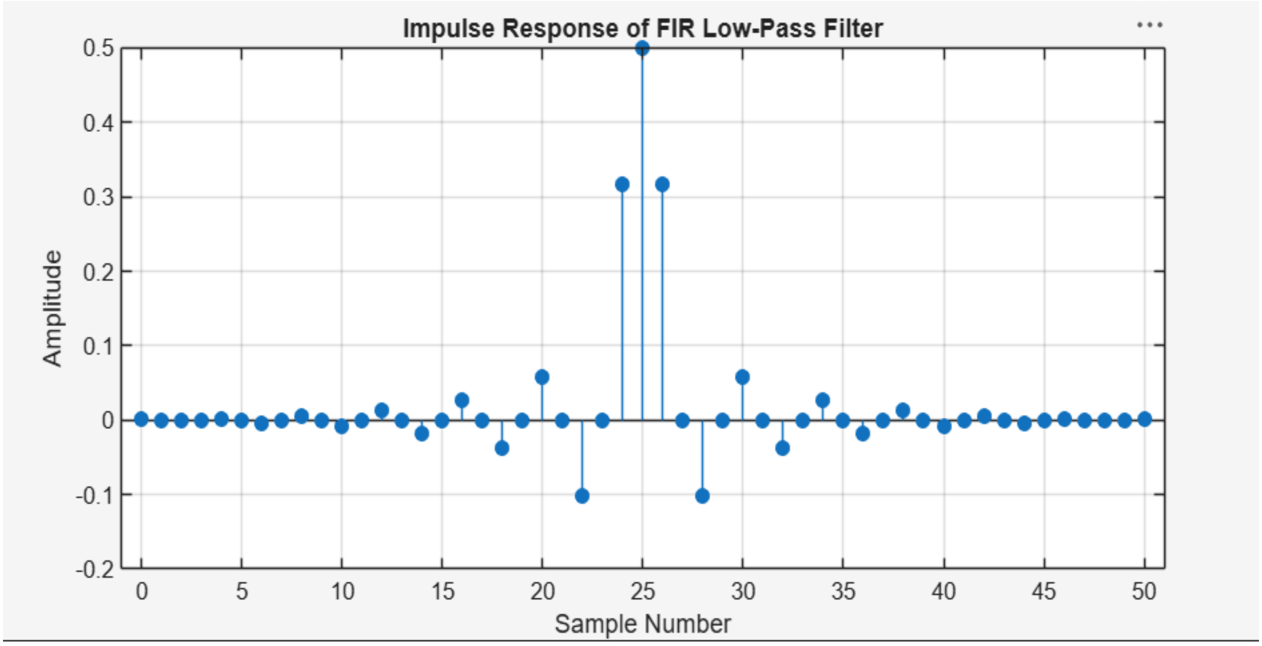
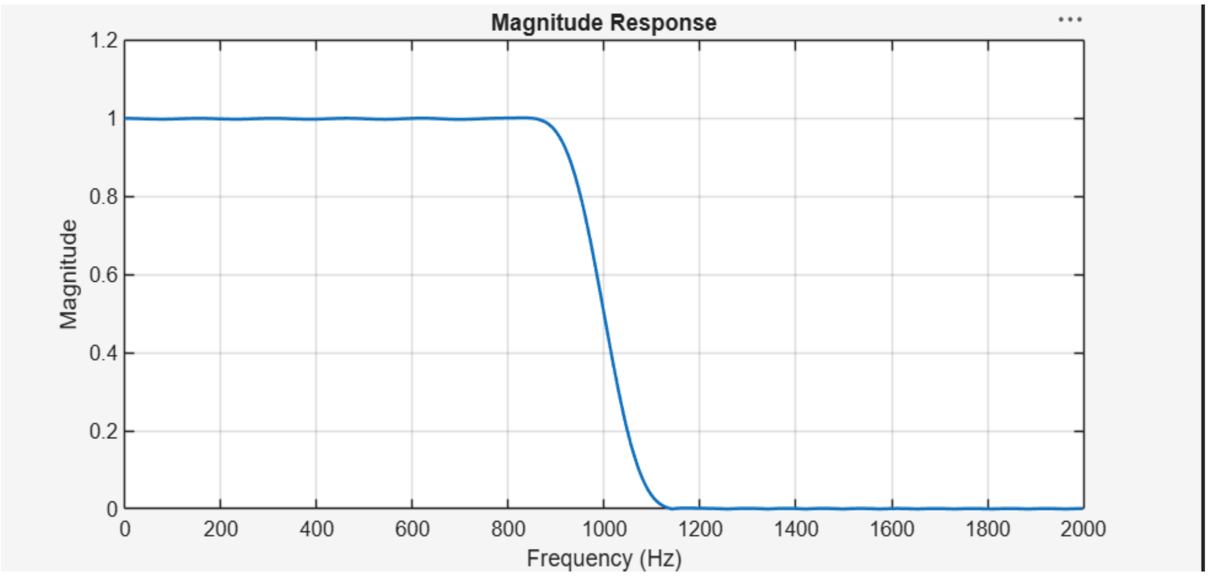

Impulse Response of FIR Low-Pass Filter
Amplitude
Sample Number
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
5
10
15
20
25
30
35
40
45
50

Fig 3:


Magnitude Response
Magnitude
Frequency (Hz)
1.2
1
0.8
0.6
0.4
0.2
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000

Expt 4	Design of Low Pass Filter using Windowing Techniques
Date:	4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

Code:

```
clc;
clear;
close all;

%% ==========================
% FIR Low-Pass Filter Design
% Using Hann (Hanning) Window
%% ==========================

% Input Parameters
fp = 1000;        % Passband cutoff frequency (Hz)
Fs = 4000;        % Sampling frequency (Hz)
N  = 50;          % Filter order

% Normalized cutoff frequency
Wn = fp/(Fs/2);

%% Design FIR Low-Pass Filter using Hann Window
b_hann = fir1(N, Wn, 'low', hann(N+1));

%% Display Filter Information
disp('========================================');
disp(' FIR Low-Pass Filter (Hann Window)');
disp('========================================');
fprintf('Filter Order         : %d\n', N);
fprintf('Sampling Frequency   : %d Hz\n', Fs);
fprintf('Cutoff Frequency     : %d Hz\n', fp);
fprintf('Normalized Cutoff    : %.2f\n\n', Wn);

disp('Filter Coefficients:');
disp(b_hann');

%% Frequency Response
figure;
freqz(b_hann,1,1024,Fs);
title('Frequency Response of FIR Low-Pass Filter (Hann Window)');

%% Impulse Response
figure;
```

```
stem(0:N,b_hann,'filled');
grid on;
xlabel('Sample Number');
ylabel('Amplitude');
title('Impulse Response of FIR Low-Pass Filter');

%% Magnitude Response
[H,f] = freqz(b_hann,1,1024,Fs);

figure;
plot(f,abs(H),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Magnitude Response');
xlim([0 Fs/2]);
```

Output:

Fig 1:

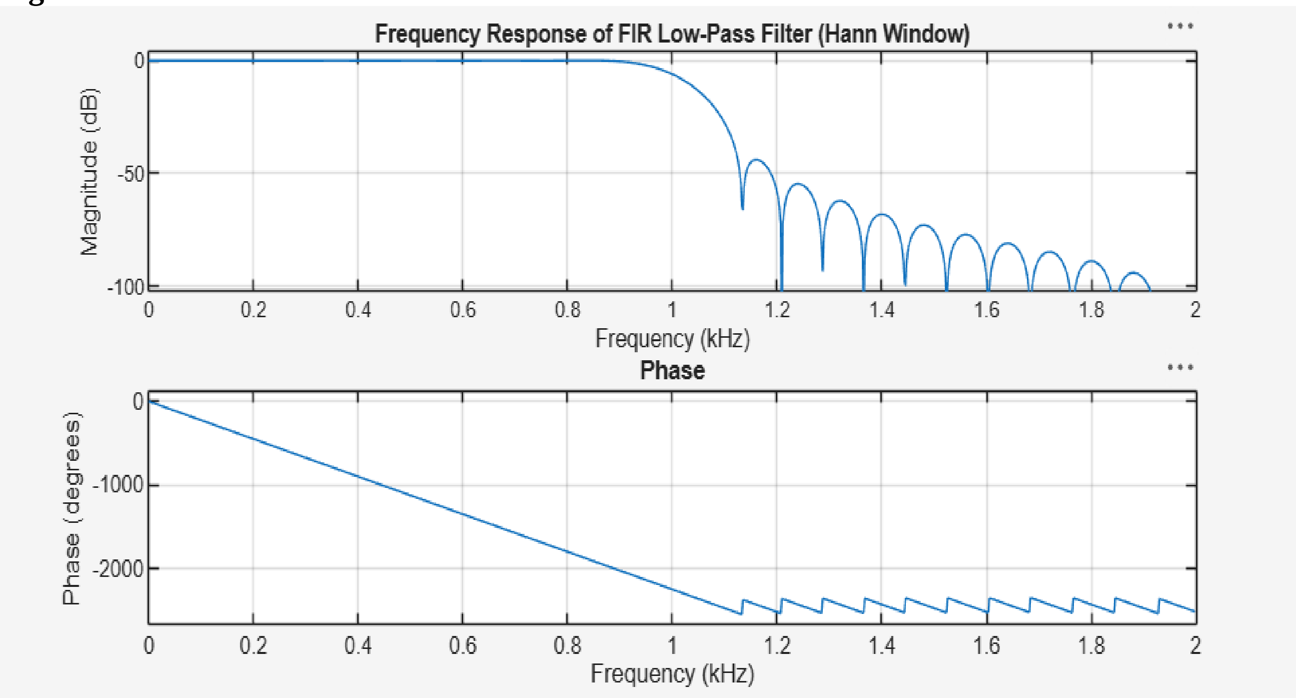
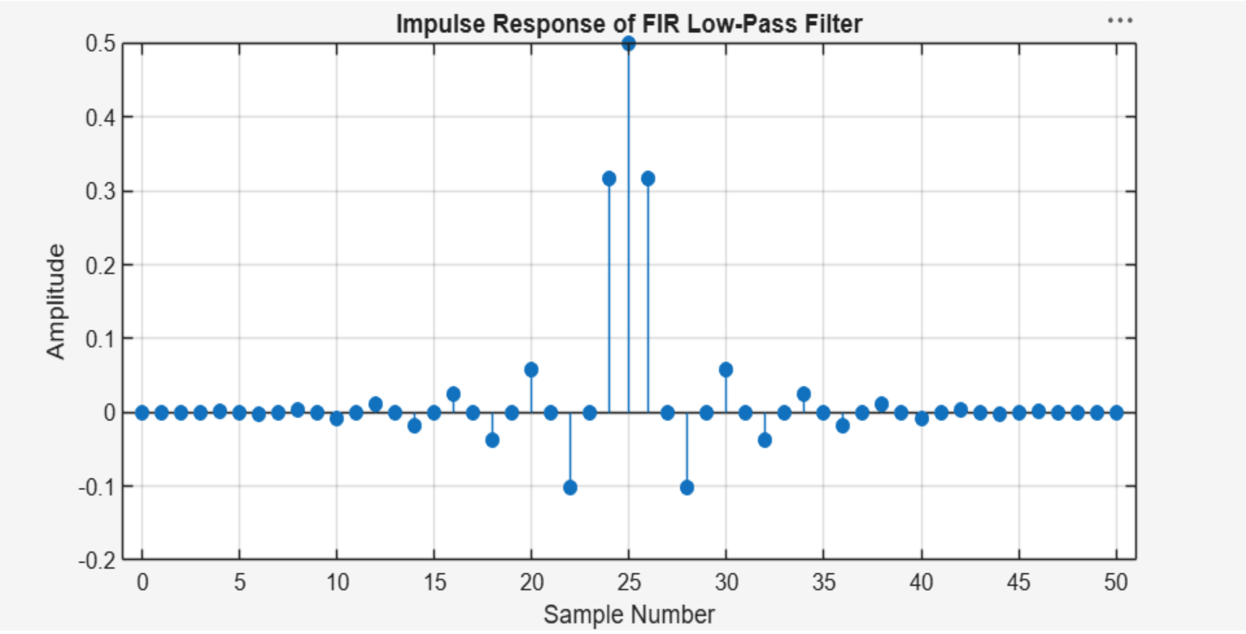
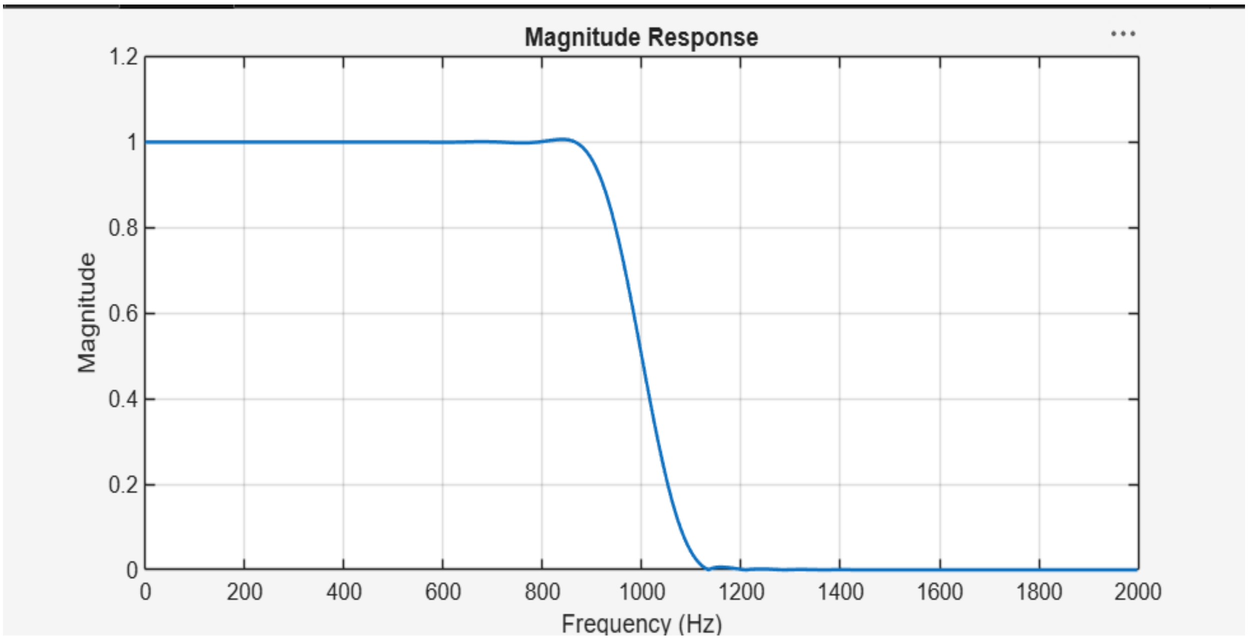


Fig 2:



Impulse Response of FIR Low-Pass Filter
Amplitude
Sample Number
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
5
10
15
20
25
30
35
40
45
50

Fig 3:



Magnitude Response
Magnitude
Frequency (Hz)
1.2
1
0.8
0.6
0.4
0.2
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000